

THEORY AND METHODS

A BULK INPUT QUEUE WITH A STAND-BY

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Summary: This paper studies the time-dependent and the steady state behaviour of a queueing system with a single channel which is subject to random failures. There is a stand-by which is employed during the period for which the regular service channel remains under repairs. The failure times as well as the repair times are exponential. The probability generating functions of the number in the system are obtained and the corresponding steady state results are derived from the time-dependent results. Two particular cases are discussed.

1. INTRODUCTION

Bulk queueing systems have been studied by several authors including Jaiswal (1960), Medhi (1975), Neuts (1967) and Chaudhry and Templeton (1983). The present paper considers a single server queueing system with batch arrivals and single departures. However, it is assumed that the service of units is interrupted due to failure of the service channel from time to time. Queueing systems which are subject to occasional interruptions have previously been studied. Among some references relating to such studies see Madan (1973, 1976, 1992). The present paper is one in that direction but considers a system which has, apart from the regular channel, the provision of a stand-by which is employed only during the repair times of the regular service channel. The stand-by may not be as efficient as the main service channel, but still may contribute a lot to avoid the queue becoming out of bounds during the failure times of the main service channel. Such situations are not uncommon. The following assumptions describe the mathematical model:

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- (i) Units (customers) arrive at the system in batches of variable size. Let $\lambda c_i dt$ ($i = 1, 2, 3, \dots$) denote the first-order probability that a batch of i units arrives in the short interval of time dt , where $0 \leq c_i \leq 1$ and $\sum_{i=1}^{\infty} c_i = 1$ and $\lambda > 0$ is the mean arrival rate of batches and the batches are pre-ordered for service purposes.
- (ii) There is a regular service channel which provides exponential service to the units one by one on a first come, first served basis with mean service time $1/\mu$ ($\mu > 0$).
- (iii) The regular service channel is subject to random failures from time to time. The operative times (times for which the regular service channel remains in working order) are assumed to be exponential with mean operative time $1/\xi$ ($\xi > 0$). Thus ξdt is the first-order probability that the regular service channel will fail during the short interval of time dt .
- (iv) As soon as the regular service channel fails, it instantly undergoes a repair process which is assumed to be exponential with mean repair time $1/\eta$ ($\eta > 0$). Thus ηdt is the first-order probability that the repairs will be completed in the short interval of time dt .
- (v) The moment the regular service channel fails it is replaced by a stand-by which keeps providing service to the units until the moment the repairs on the regular service channel are complete.
- (vi) The service times of the stand-by are assumed to be exponential with mean service time $1/\nu$ ($0 < \nu < \mu$).

2. TIME-DEPENDENT EQUATIONS GOVERNING THE SYSTEM

Define $W_n(t)$ = the probability that at time t there are n ($n \geq 0$) units in the system, including one in service, if any and the regular service channel is in working order. $R_n(t)$ = the probability that at time t there are n ($n \geq 0$)

units in the system, including one in service, if any; the regular service channel is under repairs and the stand-by is providing service. The following equations govern the system:

$$(1) \quad \frac{d}{dt} W_0(t) + (\lambda + \xi) W_0(t) = \mu W_1(t) + \eta R_0(t)$$

$$(2) \quad \frac{d}{dt} W_n(t) + (\lambda + \mu + \xi) W_n(t) = \sum_{i=1}^n \lambda c_i W_{n-i}(t) + \mu W_{n+1}(t) + \eta R_n(t) \quad (n > 0)$$

$$(3) \quad \frac{d}{dt} R_0(t) + (\lambda + \eta) R_0(t) = \nu R_1(t) + \xi W_0(t)$$

$$(4) \quad \frac{d}{dt} R_n(t) + (\lambda + \nu + \eta) R_n(t) = \sum_{i=1}^n \lambda c_i R_{n-i}(t) + \nu R_{n+1}(t) + \xi W_n(t) \quad (n > 0)$$

We assume that initially the system starts when there is no unit either waiting in the queue or being served and the service channel, though in working order, is idle so that

$$(5) \quad W_0(0) = 1$$

Let $\bar{f}(s)$ denote the Laplace transform of a function $f(t)$. Then taking Laplace transform of equations (1) to (4) and using (5), we have

$$(6) \quad (s + \lambda + \xi) \bar{W}_0(s) = 1 + \mu \bar{W}_1(s) + \eta \bar{R}_0(s)$$

$$(7) \quad (s + \lambda + \mu + \xi) \bar{W}_n(s) = \sum_{i=1}^n \lambda c_i \bar{W}_{n-i}(s) + \mu \bar{W}_{n+1}(s) + \eta \bar{R}_n(s) \quad (n > 0)$$

$$(8) \quad (s + \lambda + \eta) \bar{R}_0(s) = \nu \bar{R}_1(s) + \xi \bar{W}_0(s)$$

$$(9) \quad (s + \lambda + \nu + \eta) \bar{R}_n(s) = \sum_{i=1}^n \lambda c_i \bar{R}_{n-i}(s) + \nu \bar{R}_{n+1}(s) + \xi \bar{W}_n(s) \quad (n > 0)$$

3. THE PROBABILITY GENERATING FUNCTIONS OF THE NUMBER IN THE SYSTEM

Let $\bar{W}(s, \alpha)$ and $\bar{R}(s, \alpha)$ denote the Laplace transforms of the probability generating functions of $W_n(t)$ and $R_n(t)$ respectively. Using standard techniques, equations (6) to (9) yield

$$(10) \quad \bar{W}(s, \alpha) = \frac{\{\alpha + \mu(\alpha - 1)\bar{W}_0(s)\} \{[s + \lambda + \nu + \eta - \lambda C(\alpha)]\alpha - \nu\} + \eta \nu \alpha(\alpha - 1)\bar{R}_0(s)}{\{[s + \lambda + \mu + \xi - \lambda C(\alpha)]\alpha - \mu\} \{[s + \lambda + \nu + \eta - \lambda C(\alpha)]\alpha - \nu\} - \xi \eta \alpha^2}$$

$$(11) \quad \bar{R}(s, \alpha) = \frac{\{\nu(\alpha - 1)\bar{R}_0(s)\} \{[s + \lambda + \mu + \xi - \lambda C(\alpha)]\alpha - \mu\} + \xi \alpha \{\alpha + \mu(\alpha - 1)\bar{W}_0(s)\}}{\{[s + \lambda + \mu + \xi - \lambda C(\alpha)]\alpha - \mu\} \{[s + \lambda + \nu + \eta - \lambda C(\alpha)]\alpha - \nu\} - \xi \eta \alpha^2}$$

Since $\bar{R}(s, \alpha)$ is regular inside the contour $|\alpha| = 1$, its numerator must vanish for each of the two zeroes of the denominator giving rise to two equations in terms of the two unknowns $\bar{W}_0(s)$ and $\bar{R}_0(s)$. These two equations are sufficient to determine both these unknowns. Hence $\bar{R}(s, \alpha)$ and, for that matter $\bar{W}(s, \alpha)$ can be completely determined.

4. PROBABILITIES OF THE STATES OF THE SYSTEM

To find the probabilities of the states of the system, that is whether the service channel is in working order or under repairs at any time t , we denote these probabilities by $W(t)$ and $R(t)$ respectively and let $\alpha = 1$ in equations (10) and (11). Noting the fact that $C(1) = 1$, and inverting the resulting transforms, we have

$$(12) \quad W(t) = \frac{1}{(\xi + \eta)} \{ \eta + \xi e^{-(\xi + \eta)t} \}$$

$$(13) \quad R(t) = \frac{1}{(\xi + \eta)} \{ \xi - \xi e^{-(\xi + \eta)t} \}$$

5. THE STEADY STATE RESULTS

We define W_n and R_n as the steady state probabilities corresponding to $W_n(t)$ and $R_n(t)$ respectively and let $W(\alpha)$ and $R(\alpha)$ denote the corresponding steady state probability generating functions. Then the steady state results can be obtained by using the well-known Tauberian property

$$(14) \quad \lim_{s \rightarrow 0} s \tilde{f}(s) = \lim_{t \rightarrow \infty} f(t), \text{ if the limit on the right exists.}$$

The corresponding steady state probabilities for the two states of the system can be obtained from equations (12) and (13) on letting $t \rightarrow \infty$. Thus we have

$$(15) \quad W = \frac{\eta}{(\xi + \eta)} \quad R = \frac{\xi}{(\xi + \eta)}$$

which is intuitively true as well.

6. TWO PARTICULAR CASES

Case I

If arrivals occur one by one then $c_1 = 1$ and $c_i = 0$ for $i \neq 1$ and therefore $C(\alpha) = \alpha$. Substituting in equations (10) and (11) and performing standard calculations, we can derive the steady state results for this case. And further, we have

$$(16) \quad W_0 = \frac{\eta(\mu - \lambda)}{\mu(\xi + \eta)} \quad \text{and} \quad R_0 = \frac{\xi(\nu - \lambda)}{\nu(\xi + \eta)}$$

The steady state probability Q that there is no unit in the system irrespective of the fact whether the service channel is in working or in the failed state is given by the sum of W_0 and R_0 . Thus

$$(17) \quad Q = \frac{1}{\xi + \eta} \left\{ \eta \left(1 - \frac{\lambda}{\mu} \right) + \xi \left(1 - \frac{\lambda}{\nu} \right) \right\}$$

and the utilization factor ρ , of the system is given by $1 - Q$.

Let L_W and L_R respectively denote the average number in the system in the steady state when the service channel is in working order and when it is under repairs. Then we have

$$(18) \quad L_W = \frac{\lambda \eta}{\xi(\nu - \lambda) + \eta(\mu - \lambda)}; \quad L_R = \frac{\lambda \xi}{\xi(\nu - \lambda) + \eta(\mu - \lambda)}$$

Case II

In addition to the conditions of case I, if the system is not subject to failures, then $R_n(t) = 0$ and for that matter $\bar{R}_n(s) = 0$ for all n and $\bar{R}(s, \alpha) = 0$. Then letting $\xi = 0$, $c_i = 0$ for $i \neq 1$, $c_1 = 1$ and $C(\alpha) = \alpha$ in equation (10), we have

$$(19) \quad \bar{W}(s, \alpha) = \frac{\alpha + \mu(\alpha - 1) \bar{W}_0(s)}{\{[s + \lambda + \mu - \lambda \alpha] \alpha - \mu\}}$$

Equation (19) agrees with a known result (see Saaty, 1961, page 88, equation 4 -6 for $i = 1$).

The steady state results in this case can be obtained by letting $\xi = 0$ in the results of case I. Further, on letting $\xi = 0$ equations (17) and (18) give

$$(20) \quad Q = 1 - \frac{\lambda}{\mu}; \quad L = \frac{\lambda}{\mu - \lambda}$$

The results in equation (20) are the well-known results of the system M/M/1.

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